Nekkalapu Harichand

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Objective

Electronics and Communication Engineering graduate with hands-on experience in RF circuit design, antenna simulation, and lab validation (VNA, spectrum analyzer, signal generator), gained through a DRDO internship and academic projects. Seeking an RF / Communication Engineer role to apply this foundation to UAV communication, antenna, and sensing systems.

Education

Amrita Vishwa Vidyapeetham **2022 – 2026**
B.Tech, Electronics and Communication Engineering (Minor: AI & ML), ***CGPA: 8.15*** *Bengaluru, Karnataka*

Sasi Junior College **2020 – 2022**
Pre University, ***Percentage: 88.3%*** *Eluru, Andhra Pradesh*

Professional Experience

CABS, DRDO **Dec 2025 – Jan 2026**
Project Intern *Bengaluru, India*

- Designed a reconfigurable FPGA-based blanking management architecture to protect passive RF receivers from interference caused by multiple active transmitters, replacing rigid hardware-based blanking approaches.
- Architected the system around an Active-Passive dependency matrix stored in on-chip Block RAM (BRAM), initialized via a MATLAB-generated COE file through Xilinx's block memory generator, enabling system reconfiguration without RTL redesign.
- Verified functional correctness through RTL simulation in Xilinx Vivado, validating deterministic OR-based blanking decisions, BRAM initialization, and scalable 32-bit active/passive subsystem selection at a 100 MHz system clock.

Projects

Bandpass Filter Design | *ADS, HFSS, VNA, Signal Generator, Spectrum Analyzer* **June 2025 – Nov 2025**

- Designed a parallel-coupled microstrip bandpass filter centered at 6.4 GHz with a 300 MHz bandwidth using ADS and HFSS, deriving even/odd-mode characteristic impedances from filter theory to meet target S-parameter specifications.
- Applied vector fitting to extract a rational state-space model (poles and residues) from simulated S21 data, achieving a final RMS fitting error of 1.36×10^{-4} to validate and refine the design before fabrication.
- Fabricated the filter as a PCB on a Kappa 431 substrate with SMA connectors and validated using a VNA, signal generator, and spectrum analyzer, measuring a 6.4 GHz center frequency with -2.6 dB insertion loss consistent with simulation.

Efficient Estimation of Soil Moisture Using Bi-Static GPR | *gprMax, MATLAB, HFSS* **July 2025 – May 2026**

- Designed a horn antenna and simulated 2-layer and 3-layer configurations in bistatic and monostatic setups using HFSS to characterize subsurface radar response.
- Simulated GPR responses using gprMax (FDTD) and implemented Green's Function-based full waveform inversion for subsurface characterization.
- Estimated dielectric permittivity and conductivity using PSO, GA, and fminsearch optimization algorithms, applying Topp's Equation and CRIM for soil moisture estimation.

FPGA-Based Error Correction Techniques | — *Verilog, Xilinx Vivado* **Sept 2024– Nov 2024**

- Designed and implemented three cascading ECC schemes in Verilog HDL– Hamming (7,4), SEC-DED (8,4), and an extended SEC-DED-DAEC (14,8)– supporting single-error correction, double-error detection, and double-adjacent-error correction for SRAM-based memory protection.
- Built syndrome-based decoding logic with a lookup-table approach for the SEC-DED-DAEC scheme, verified through testbench simulation with targeted single-bit, double-adjacent, and double-non-adjacent error injection.
- Synthesized and implemented all three designs on a Basys-3 (Artix-7) FPGA via Xilinx Vivado, confirming zerotiming violations and low LUT/FF utilization across all variants.

Technical Skills

Coding: C, Python, Verilog, SystemVerilog
Tools: LTSpice, MATLAB, ADS, HFSS, Xilinx Vivado, VNA, Signal Generator, Spectrum Analyzer, Linux, Git
Machine Learning: TensorFlow, Scikit-learn, NumPy, Pandas

Certifications

Drone Engineering – Mentorship Program, Teachnook (in association with Drona Aviation) **Jul 2024 – Aug 2024**

- Completed a mentorship program covering the fundamentals of drone design and systems. Certificate ID: TNK2404-074980.

Achievements

- **Paper Presentation** – Presented a research paper at IEEE SPACE 2026 (3rd edition – Space, Aerospace and defenCE Conference), organized by IEEE AESS, Bangalore, July 2026.
- **Treasurer, ECCF** – Organized club events and conducted hackathons, alongside managing club finances and budgeting.